#1 · Cardiothoracic

Which chest radiograph border is lost in right middle lobe disease?

Answer

The right heart border, by the silhouette sign. Lingular disease instead obscures the left heart border.

#2 · Cardiothoracic

What does loss of the hemidiaphragm outline localise disease to?

Answer

The lower lobe, because the lower lobe abuts the diaphragm. Right middle lobe and lingular disease spare it.

#3 · Cardiothoracic

What is the cervicothoracic sign?

Answer

A mass whose edge remains visible above the clavicle lies in the posterior mediastinum. A mass whose edge disappears at the clavicle is anterior.

#4 · Cardiothoracic

What does an air bronchogram indicate about the proximal airway?

Answer

The bronchus is patent and alveoli are filled with fluid, pus, cells or blood. It argues against proximal obstructive collapse.

#5 · Cardiothoracic

What is the Golden S sign?

Answer

Right upper lobe collapse around a central mass: the elevated concave horizontal fissure joins the convex hilar mass to form an S shape. It suggests a central obstructing carcinoma.

#6 · Cardiothoracic

What is the Luftsichel sign and when does it occur?

Answer

A crescent of lucency beside the aortic arch in left upper lobe collapse, caused by the hyperinflated superior segment of the left lower lobe.

#7 · Cardiothoracic

How does mediastinal shift distinguish collapse from effusion in an opaque hemithorax?

Answer

Shift toward the opaque side suggests collapse, shift away suggests a large effusion, and no shift suggests consolidation or mixed disease.

#8 · Cardiothoracic

What is the signet ring sign?

Answer

A dilated bronchus wider than its accompanying pulmonary artery on HRCT, diagnostic of bronchiectasis.

#9 · Cardiothoracic

Which bronchiectasis distribution suggests cystic fibrosis or ABPA?

Answer

Upper zone and central predominance. In ABPA, mucoid impaction produces finger-in-glove opacities that may be of high attenuation.

#10 · Cardiothoracic

Which emphysema pattern is lower zone and linked to alpha-1 antitrypsin deficiency?

Answer

Panlobular emphysema. Centrilobular emphysema is upper zone and linked to smoking.

#11 · Cardiothoracic

What CT distribution of nodules suggests sarcoidosis or lymphangitic carcinomatosis?

Answer

Perilymphatic nodules along the pleura, fissures and bronchovascular bundles. Centrilobular nodules spare the pleura, and random nodules suggest haematogenous spread.

#12 · Cardiothoracic

What does a tree-in-bud pattern indicate?

Answer

Endobronchial spread of infection, such as tuberculosis, non-tuberculous mycobacteria or bronchiolitis, within centrilobular small airways.

#13 · Cardiothoracic

What are the classic HRCT features of a UIP pattern?

Answer

Basal, subpleural predominant reticulation with honeycombing and traction bronchiectasis, with little ground glass. In the right clinical setting biopsy is unnecessary.

#14 · Cardiothoracic

Which HRCT feature separates NSIP from UIP?

Answer

Basal ground glass and reticulation with subpleural sparing and minimal honeycombing. NSIP is typical of connective tissue disease.

#15 · Cardiothoracic

What is the Garland triad in sarcoidosis?

Answer

Bilateral hilar plus right paratracheal lymphadenopathy, also called the 1-2-3 sign.

#16 · Cardiothoracic

What are the radiographic stages of sarcoidosis?

Answer

Stage 0 normal, I lymphadenopathy alone, II lymphadenopathy with parenchymal disease, III parenchymal disease alone, IV fibrosis.

#17 · Cardiothoracic

Which CT sign suggests organising pneumonia, and what else can cause it?

Answer

The reversed halo (atoll) sign: ring of consolidation around central ground glass. It also occurs in pulmonary infarct and mucormycosis.

#18 · Cardiothoracic

Where do asbestos-related pleural plaques lie and what do they spare?

Answer

Over the diaphragm and lower chest wall, often calcified, sparing the costophrenic angles and apices.

#19 · Cardiothoracic

Which pneumonia shows a bulging fissure?

Answer

Klebsiella pneumoniae, which also tends to cavitate and is associated with alcohol misuse.

#20 · Cardiothoracic

What is the radiographic pattern of Pneumocystis jirovecii pneumonia?

Answer

Bilateral perihilar ground glass with relative peripheral sparing, with thin-walled cysts in about a third that predispose to pneumothorax.

#21 · Cardiothoracic

Where does post-primary tuberculosis characteristically occur?

Answer

The apical and posterior segments of the upper lobes and the superior segments of the lower lobes, with cavitation and tree-in-bud spread.

#22 · Cardiothoracic

What is the Ranke complex?

Answer

A calcified Ghon focus with a calcified ipsilateral hilar lymph node, the healed result of primary tuberculosis.

#23 · Cardiothoracic

What features separate cardiogenic pulmonary oedema from ARDS early on?

Answer

Cardiogenic oedema has cardiomegaly, septal lines, upper lobe venous diversion and effusions. ARDS has a normal heart size and diffuse bilateral opacity.

#24 · Cardiothoracic

What volume of pleural fluid blunts the costophrenic angle on an erect PA film?

Answer

About 200 mL, although about 50 mL blunts the posterior angle on the lateral film.

#25 · Cardiothoracic

What are Light criteria for an exudate?

Answer

Pleural to serum protein above 0.5, pleural to serum LDH above 0.6, or pleural LDH above two thirds of the upper limit of normal serum LDH.

#26 · Cardiothoracic

How does empyema differ from a lung abscess on CT?

Answer

Empyema forms an obtuse angle with the chest wall, compresses lung and shows the split pleura sign. An abscess forms acute angles and has a thick irregular wall.

#27 · Cardiothoracic

What is the deep sulcus sign?

Answer

An abnormally deep lucent costophrenic angle on a supine film, indicating anterior pneumothorax.

#28 · Cardiothoracic

Which features favour mesothelioma over metastatic pleural disease?

Answer

Circumferential nodular pleural thickening, mediastinal pleural involvement and an encased, volume-reduced hemithorax.

#29 · Cardiothoracic

Which mediastinal compartment contains thymoma, lymphoma, germ cell tumour and retrosternal thyroid?

Answer

The prevascular compartment, remembered as the four Ts: thymoma, teratoma, terrible lymphoma, thyroid.

#30 · Cardiothoracic

Which paraneoplastic conditions are associated with thymoma?

Answer

Myasthenia gravis, pure red cell aplasia and hypogammaglobulinaemia.

#31 · Cardiothoracic

How can a mediastinal teratoma be distinguished from a seminoma?

Answer

A mature teratoma contains fat, fluid and calcification. A seminoma is homogeneous with a normal AFP, whereas non-seminomatous tumours are heterogeneous with a raised AFP.

#32 · Cardiothoracic

Which solid nodule size under Fleischner 2017 needs no routine follow-up in a low risk adult?

Answer

A solid nodule under 6 mm. At 6 to 8 mm, CT at 6 to 12 months is advised, then consider CT at 18 to 24 months.

#33 · Cardiothoracic

What is the Fleischner 2017 advice for a part-solid nodule of 6 mm or more?

Answer

CT at 3 to 6 months; if it persists with a solid component under 6 mm, annual CT for 5 years. A solid component of 6 mm or more is highly suspicious and needs referral.

#34 · Cardiothoracic

To whom does Fleischner 2017 not apply?

Answer

Patients under 35, those in lung cancer screening, immunosuppressed patients and those with a known primary malignancy.

#35 · Cardiothoracic

What is the follow-up for a Lung-RADS 3 result?

Answer

A 6 month low dose CT. Category 4A needs a 3 month CT, or PET/CT if the solid component is 8 mm or more.

#36 · Cardiothoracic

What are the TNM 9th edition T categories by size for lung cancer?

Answer

T1 up to 3 cm, T2 over 3 to 5 cm, T3 over 5 to 7 cm and T4 over 7 cm, with T1 and T2 subdivided in 1 cm steps.

#37 · Cardiothoracic

What changed in M1c in TNM 9th edition?

Answer

M1c was divided into M1c1 (multiple extrathoracic lesions in one organ system) and M1c2 (multiple lesions in more than one organ system).

#38 · Cardiothoracic

What distinguishes a thick-walled malignant cavity from a benign one?

Answer

A wall thicker than 15 mm favours malignancy and a wall thinner than 4 mm favours a benign cause.

#39 · Cardiothoracic

What is the Pancoast tumour presentation?

Answer

A superior sulcus tumour invading the chest wall apex, causing Horner syndrome and pain in the C8 to T1 distribution. MRI best shows brachial plexus and vessel involvement.

#40 · Cardiothoracic

Which compartment of the mediastinum do neurogenic tumours occupy?

Answer

The paravertebral compartment. Schwannoma and neurofibroma give dumbbell masses, and neuroblastoma in children may calcify.

#41 · Cardiothoracic

What late gadolinium enhancement pattern is typical of ischaemic scar?

Answer

Subendocardial or transmural enhancement that follows a coronary territory. Non-ischaemic disease usually spares the subendocardium.

#42 · Cardiothoracic

Which enhancement pattern is characteristic of dilated cardiomyopathy with worse arrhythmic prognosis?

Answer

A linear midwall stripe of enhancement in the septum, with no subendocardial enhancement.

#43 · Cardiothoracic

What wall thickness defines hypertrophic cardiomyopathy in an adult?

Answer

15 mm or more, usually asymmetric septal, in the absence of another cause such as hypertension or aortic stenosis.

#44 · Cardiothoracic

Where is late enhancement often seen in hypertrophic cardiomyopathy?

Answer

Patchy midwall enhancement, often at the right ventricular insertion points.

#45 · Cardiothoracic

Which MRI findings suggest cardiac amyloidosis?

Answer

Diffuse subendocardial or transmural enhancement not following a coronary territory, difficulty nulling the myocardium, and markedly raised native T1 and extracellular volume.

#46 · Cardiothoracic

Which scintigraphy agent is strongly positive in transthyretin cardiac amyloid?

Answer

Technetium pyrophosphate or DPD scintigraphy shows intense myocardial uptake.

#47 · Cardiothoracic

What is the typical enhancement distribution in acute myocarditis?

Answer

Subepicardial or midwall enhancement, classically inferolateral, with T2 evidence of oedema.

#48 · Cardiothoracic

How does takotsubo cardiomyopathy appear on MRI?

Answer

Apical ballooning with basal hyperkinesis, oedema and usually no significant enhancement, with normal coronary arteries.

#49 · Cardiothoracic

What T2 star value indicates myocardial iron loading?

Answer

Below 20 ms indicates iron loading, and below 10 ms indicates severe loading.

#50 · Cardiothoracic

What are the CAD-RADS 2.0 categories 3 and 4A?

Answer

Category 3 is 50 to 69 percent stenosis. Category 4A is 70 to 99 percent in one or two vessels.

#51 · Cardiothoracic

What defines CAD-RADS 4B?

Answer

Left main stenosis above 50 percent, or three vessel disease with each stenosis 70 percent or more.

#52 · Cardiothoracic

Why does dense coronary calcium limit CT angiography?

Answer

Blooming artefact overestimates stenosis, so a functional test or invasive angiography may be needed.

#53 · Cardiothoracic

Why are heart rate control and glyceryl trinitrate used before coronary CT?

Answer

A slow regular rate, usually under 60 beats per minute, reduces motion, and nitrate dilates the coronary arteries.

#54 · Cardiothoracic

Which ostium type of atrial septal defect is associated with Down syndrome?

Answer

Ostium primum, with a cleft mitral valve.

#55 · Cardiothoracic

Which atrial septal defect is associated with partial anomalous pulmonary venous return?

Answer

The sinus venosus defect, high near the superior vena cava, usually with right upper lobe veins.

#56 · Cardiothoracic

Which ventricular septal defect type is most common?

Answer

The perimembranous type, and VSD is the commonest congenital heart defect overall.

#57 · Cardiothoracic

What radiographic signs suggest aortic coarctation?

Answer

Figure of three sign at the aortic contour and inferior rib notching of the third to eighth ribs, with sparing of the first and second.

#58 · Cardiothoracic

Which congenital valve anomaly is most commonly associated with coarctation?

Answer

Bicuspid aortic valve.

#59 · Cardiothoracic

What is the radiographic appearance of tetralogy of Fallot?

Answer

A boot shaped heart with a concave pulmonary bay and oligaemic lungs, with a right aortic arch in about a quarter.

#60 · Cardiothoracic

Which congenital lesion gives an egg on a string heart?

Answer

Complete transposition of the great arteries, with a narrow vascular pedicle.

#61 · Cardiothoracic

What is the snowman heart sign?

Answer

Supracardiac total anomalous pulmonary venous connection, usually seen in later infancy.

#62 · Cardiothoracic

What calcium score thresholds favour severe aortic stenosis when echocardiography is discordant?

Answer

About 2000 Agatston units in men and 1200 in women.

#63 · Cardiothoracic

Which measurements does CT make before transcatheter aortic valve implantation?

Answer

Annulus area and perimeter, coronary ostial heights, sinus of Valsalva size, root calcification and iliofemoral access.

#64 · Cardiothoracic

What chest radiograph signs suggest rheumatic mitral stenosis?

Answer

Left atrial enlargement with a straight left heart border, double density behind the right heart border and upper lobe venous diversion.

#65 · Cardiothoracic

How does mitral annular calcification typically appear?

Answer

As a C shaped or ring shaped calcification at the posterior annulus, common in older patients, dialysis and hyperparathyroidism.

#66 · Cardiothoracic

What pericardial thickness suggests constrictive pericarditis?

Answer

Above 4 mm, often calcified, with a small tubular ventricle and dilated atria and inferior vena cava.

#67 · Cardiothoracic

How is constriction separated from restrictive cardiomyopathy?

Answer

In constriction the pericardium is thickened and the myocardium normal, with septal bounce. In restriction the pericardium is normal and the myocardium abnormal.

#68 · Cardiothoracic

Where is the typical site of a pericardial cyst?

Answer

The right cardiophrenic angle, as a thin walled non-enhancing water density lesion.

#69 · Cardiothoracic

What is the Stanford classification of aortic dissection?

Answer

Type A involves the ascending aorta and is a surgical emergency. Type B spares it and is usually managed medically unless complicated.

#70 · Cardiothoracic

How does intramural haematoma appear on non-contrast CT?

Answer

A crescentic or circumferential high attenuation wall thickening of 60 to 70 Hounsfield units, with no flap and no enhancement.

#71 · Cardiothoracic

Where does traumatic aortic injury usually occur?

Answer

At the aortic isthmus just distal to the left subclavian artery, from deceleration.

#72 · Cardiothoracic

Which CT signs indicate right heart strain in pulmonary embolism?

Answer

Right to left ventricular ratio above 1, septal bowing toward the left, dilated main pulmonary artery and contrast reflux into the inferior vena cava.

#73 · Cardiothoracic

What is the polo mint sign?

Answer

A central filling defect surrounded by contrast in an acutely occluded pulmonary artery seen in cross section.

#74 · Cardiothoracic

How does chronic thromboembolic disease differ from acute embolism on CT?

Answer

Eccentric wall adherent thrombus at an obtuse angle, webs, bands, vessel narrowing and mosaic attenuation.

#75 · Cardiothoracic

Which test is preferred for screening chronic thromboembolic pulmonary hypertension?

Answer

The ventilation perfusion scan, which has high sensitivity.

#76 · Cardiothoracic

What main pulmonary artery size suggests pulmonary hypertension?

Answer

A diameter above about 29 mm or a pulmonary artery to ascending aorta ratio above 1.

#77 · Cardiothoracic

What is the correct position of an endotracheal tube tip?

Answer

About 5 cm above the carina, with an acceptable range of 3 to 7 cm in neutral head position.

#78 · Cardiothoracic

Where should the tip of a central venous catheter lie?

Answer

At the lower superior vena cava or cavoatrial junction, above the pericardial reflection.

#79 · Cardiothoracic

Where should an intra-aortic balloon pump tip lie?

Answer

1 to 2 cm below the top of the aortic knuckle, just distal to the origin of the left subclavian artery.

#80 · Cardiothoracic

Which side is diaphragmatic rupture more often diagnosed, and why?

Answer

The left, because the liver protects the right. Look for herniated stomach or bowel and a collar sign.

#81 · Abdominal and Pelvic

Which T2 and enhancement pattern characterises hepatic haemangioma?

Answer

Very bright T2 signal close to CSF and peripheral nodular discontinuous enhancement that fills centripetally and persists on delayed images.

#82 · Abdominal and Pelvic

What discriminates focal nodular hyperplasia from adenoma on gadoxetate MRI?

Answer

Focal nodular hyperplasia is iso to hyperintense in the hepatobiliary phase; adenoma is usually hypointense because it lacks bile ducts.

#83 · Abdominal and Pelvic

Which feature of focal nodular hyperplasia is bright on T2 and enhances late?

Answer

The central scar.

#84 · Abdominal and Pelvic

Which hepatocellular adenoma subtype is diffusely fatty with signal drop on opposed phase imaging?

Answer

HNF1 alpha inactivated adenoma.

#85 · Abdominal and Pelvic

Which adenoma subtype carries the highest malignant risk?

Answer

Beta catenin activated adenoma, more common in men.

#86 · Abdominal and Pelvic

Which adenoma findings usually prompt resection or intervention?

Answer

Size over 5 cm, any adenoma in a man, and haemorrhage.

#87 · Abdominal and Pelvic

Which LI-RADS major features apply to hepatocellular carcinoma?

Answer

Non rim arterial phase hyperenhancement, nonperipheral washout, enhancing capsule and threshold growth.

#88 · Abdominal and Pelvic

What does LR-M denote?

Answer

Probable or definite malignancy that is not specific for hepatocellular carcinoma, such as a targetoid lesion typical of cholangiocarcinoma.

#89 · Abdominal and Pelvic

What does LR-TIV denote?

Answer

Tumour in vein, for example enhancing expansile portal vein tumour thrombus.

#90 · Abdominal and Pelvic

In which patients is LI-RADS valid?

Answer

Those at high risk for hepatocellular carcinoma, mainly cirrhosis or chronic hepatitis B; not a non cirrhotic liver.

#91 · Abdominal and Pelvic

What is the classic appearance of colorectal liver metastases?

Answer

Hypovascular lesions with peripheral rim enhancement and a bright T2 rim, with calcification in mucinous tumours.

#92 · Abdominal and Pelvic

Which liver metastases are typically hypervascular?

Answer

Neuroendocrine, renal cell, thyroid and melanoma metastases.

#93 · Abdominal and Pelvic

Which signs suggest a pyogenic liver abscess?

Answer

Thick enhancing wall, internal septa, the cluster sign and a double target appearance.

#94 · Abdominal and Pelvic

What is the water lily sign?

Answer

A floating collapsed membrane in a hydatid cyst, usually with daughter cysts and a calcified wall.

#95 · Abdominal and Pelvic

How do you separate focal fat sparing from a mass?

Answer

Fat has no mass effect or vessel distortion, sits beside the gallbladder fossa or portal bifurcation, and drops signal on opposed phase MRI.

#96 · Abdominal and Pelvic

How does iron overload behave on gradient echo MRI compared with fat?

Answer

Signal loss increases with longer echo time on in phase images, whereas fat drops on opposed phase.

#97 · Abdominal and Pelvic

Which signs of portal hypertension are seen on imaging?

Answer

Portal vein over 13 mm, splenomegaly and collaterals around the stomach, oesophagus, umbilical vein and splenorenal region.

#98 · Abdominal and Pelvic

What does Budd-Chiari syndrome show on imaging?

Answer

Hepatic venous outflow obstruction with enlarged caudate, patchy central enhancement and comma shaped intrahepatic collaterals.

#99 · Abdominal and Pelvic

How does primary haemochromatosis differ from transfusional iron overload?

Answer

Primary spares the spleen; transfusional overload also darkens spleen and marrow.

#100 · Abdominal and Pelvic

What is the most specific ultrasound sign of acute cholecystitis?

Answer

A stone impacted in the gallbladder neck, with wall over 3 mm, pericholecystic fluid and sonographic Murphy sign.

#101 · Abdominal and Pelvic

What does non filling of the gallbladder at 60 minutes on HIDA suggest?

Answer

Acute cholecystitis from cystic duct obstruction.

#102 · Abdominal and Pelvic

What is emphysematous cholecystitis?

Answer

Gas in the gallbladder wall or lumen, typically in diabetic patients, with high risk of perforation.

#103 · Abdominal and Pelvic

What is the ultrasound hallmark of adenomyomatosis?

Answer

Intramural cysts with comet tail artefacts from Rokitansky-Aschoff sinuses.

#104 · Abdominal and Pelvic

What gallbladder polyp size usually prompts surgical referral?

Answer

10 mm or more.

#105 · Abdominal and Pelvic

What is the double duct sign and what does it suggest?

Answer

Dilated common bile duct and pancreatic duct, suggesting a pancreatic head or ampullary tumour.

#106 · Abdominal and Pelvic

Which modality best assesses choledocholithiasis without radiation?

Answer

MRCP.

#107 · Abdominal and Pelvic

What is the typical appearance of primary sclerosing cholangitis?

Answer

Multifocal short strictures and dilatation giving a beaded duct appearance, associated with ulcerative colitis and cholangiocarcinoma.

#108 · Abdominal and Pelvic

Which Todani type is Caroli disease?

Answer

Type V, intrahepatic ductal dilatation with the central dot sign.

#109 · Abdominal and Pelvic

Which system classifies perihilar cholangiocarcinoma extent?

Answer

Bismuth-Corlette.

#110 · Abdominal and Pelvic

When should CT be done in acute pancreatitis?

Answer

Only when diagnosis is uncertain or complications are suspected, ideally after 72 hours.

#111 · Abdominal and Pelvic

How does walled off necrosis differ from a pseudocyst?

Answer

Both are beyond 4 weeks with a wall; walled off necrosis contains solid debris, pseudocyst only fluid.

#112 · Abdominal and Pelvic

Which finding on CT suggests infected pancreatic necrosis?

Answer

Gas within the collection.

#113 · Abdominal and Pelvic

Which vascular complications occur in pancreatitis?

Answer

Splenic vein thrombosis and pseudoaneurysm of the splenic or gastroduodenal artery.

#114 · Abdominal and Pelvic

What is the CT appearance of autoimmune pancreatitis type 1?

Answer

Sausage shaped diffusely enlarged gland with a non enhancing capsule like rim, raised IgG4 and steroid response.

#115 · Abdominal and Pelvic

What vascular contact makes pancreatic adenocarcinoma borderline resectable versus unresectable?

Answer

SMA or coeliac contact of 180 degrees or less is borderline; over 180 degrees is unresectable.

#116 · Abdominal and Pelvic

Which pancreatic cyst is microcystic with a central scar and sunburst calcification?

Answer

Serous cystadenoma.

#117 · Abdominal and Pelvic

What are high risk stigmata of IPMN?

Answer

Enhancing mural nodule of 5 mm or more, main duct 10 mm or more, or obstructive jaundice with a cystic lesion in the pancreatic head.

#118 · Abdominal and Pelvic

What is the CT appearance of a splenic infarct?

Answer

A peripheral wedge shaped low attenuation area.

#119 · Abdominal and Pelvic

Which splenic injury findings favour embolisation?

Answer

Active contrast extravasation or a pseudoaneurysm.

#120 · Abdominal and Pelvic

What are the features of Crohn disease on CT or MR enterography?

Answer

Segmental asymmetric thickening, comb sign, creeping fat, skip lesions, fistulae and abscesses.

#121 · Abdominal and Pelvic

How does ulcerative colitis differ from Crohn disease on imaging?

Answer

Continuous symmetrical disease from the rectum, loss of haustra (lead pipe colon) and pseudopolyps.

#122 · Abdominal and Pelvic

Where does ischaemic colitis typically occur?

Answer

Watershed regions: splenic flexure and rectosigmoid junction.

#123 · Abdominal and Pelvic

What defines small bowel obstruction on CT?

Answer

Small bowel over 3 cm with a transition point and collapsed distal bowel.

#124 · Abdominal and Pelvic

Which CT signs suggest bowel ischaemia in obstruction?

Answer

Mesenteric swirl, reduced wall enhancement, wall thickening, pneumatosis, portal venous gas and free fluid.

#125 · Abdominal and Pelvic

What is Rigler triad?

Answer

Small bowel obstruction, pneumobilia and an ectopic gallstone in gallstone ileus.

#126 · Abdominal and Pelvic

What are normal maximum bowel diameters on an abdominal radiograph?

Answer

Small bowel 3 cm, colon 6 cm, caecum 9 cm.

#127 · Abdominal and Pelvic

Which sonographic findings suggest appendicitis?

Answer

A non compressible appendix over 6 mm, appendicolith and periappendiceal fat stranding.

#128 · Abdominal and Pelvic

How is diverticulitis graded by modified Hinchey?

Answer

Ia phlegmon, Ib pericolic abscess, II pelvic or distant abscess, III purulent peritonitis, IV faecal peritonitis.

#129 · Abdominal and Pelvic

Which contrast is used when perforation is suspected?

Answer

Water soluble low osmolar contrast, as barium causes peritonitis or mediastinitis.

#130 · Abdominal and Pelvic

What are the signs of intussusception on ultrasound?

Answer

Target sign in transverse and pseudokidney sign in longitudinal section.

#131 · Abdominal and Pelvic

What is the Bosniak 2019 feature that separates Class III from Class IV?

Answer

Class IV has an enhancing nodule (convex protrusion 4 mm or more with obtuse margins, or any size with acute margins). Class III has thick (4 mm or more) or irregular enhancing walls or septa without a nodule.

#132 · Abdominal and Pelvic

What approximate malignancy risks are attached to Bosniak III and IV?

Answer

About 50 per cent for Class III and about 90 per cent for Class IV.

#133 · Abdominal and Pelvic

What CT change in attenuation defines enhancement of a renal lesion?

Answer

An increase of 20 HU or more between unenhanced and post-contrast images. A change under 10 HU is non-enhancement.

#134 · Abdominal and Pelvic

Which Bosniak class has four or more smooth thin enhancing septa or smooth minimal wall thickening of 3 mm or less?

Answer

Class IIF, which is followed by imaging.

#135 · Abdominal and Pelvic

What finding makes a renal mass an angiomyolipoma?

Answer

Macroscopic fat, below minus 20 HU on CT or signal loss on fat-suppressed MRI.

#136 · Abdominal and Pelvic

When does angiomyolipoma bleeding risk rise, and what treatment is considered?

Answer

Risk rises with size above 4 cm and intralesional aneurysms of 5 mm or more. Embolisation is considered.

#137 · Abdominal and Pelvic

What is the typical enhancement pattern of clear cell renal cell carcinoma?

Answer

Avid corticomedullary enhancement, heterogeneity with necrosis, and washout in the nephrographic phase.

#138 · Abdominal and Pelvic

How does papillary renal carcinoma differ from clear cell on imaging?

Answer

It is hypovascular with weak enhancement and low T2 signal.

#139 · Abdominal and Pelvic

Why is a central scar not reliable for diagnosing oncocytoma?

Answer

Central stellate scars also occur in renal cell carcinoma, so tissue diagnosis or resection is still needed.

#140 · Abdominal and Pelvic

What is the T1a, T1b, T2a and T2b size ladder for renal cell carcinoma?

Answer

T1a 4 cm or less, T1b more than 4 to 7 cm, T2a more than 7 to 10 cm, T2b more than 10 cm, all confined to the kidney.

#141 · Abdominal and Pelvic

What defines T3b and T3c renal cell carcinoma?

Answer

T3b is tumour thrombus in the IVC below the diaphragm. T3c is thrombus above the diaphragm or invasion of the IVC wall.

#142 · Abdominal and Pelvic

How can tumour thrombus be told from bland thrombus?

Answer

Tumour thrombus enhances and expands the vein. Bland thrombus does not enhance.

#143 · Abdominal and Pelvic

What is the first-line imaging test for suspected ureteric colic in an adult?

Answer

Unenhanced low-dose CT of the kidneys, ureters and bladder.

#144 · Abdominal and Pelvic

Which imaging test is preferred for suspected renal colic in pregnancy?

Answer

Ultrasound first, with MRI as the problem-solving test.

#145 · Abdominal and Pelvic

What sign helps separate a ureteric stone from a pelvic phlebolith?

Answer

The soft-tissue rim sign of oedematous ureteral wall around the stone. A phlebolith is round with central lucency and a comet tail.

#146 · Abdominal and Pelvic

At what size do ureteric stones usually pass spontaneously?

Answer

Stones of 5 mm or less pass in most patients. Passage becomes unlikely above 10 mm.

#147 · Abdominal and Pelvic

How do uric acid stones appear on plain film and CT?

Answer

Radiolucent on plain film but dense on CT, usually 200 to 500 HU.

#148 · Abdominal and Pelvic

Which stone type is not visible on CT?

Answer

Indinavir stones, recognised through secondary signs such as hydronephrosis.

#149 · Abdominal and Pelvic

Which condition produces staghorn calculi with alkaline urine?

Answer

Struvite stones from urease-producing organisms.

#150 · Abdominal and Pelvic

What does diuretic renography (MAG3) show in true obstruction?

Answer

A rising curve with delayed washout after furosemide, with T1/2 above 20 minutes.

#151 · Abdominal and Pelvic

What is the Weigert-Meyer rule?

Answer

In a duplex system the upper pole ureter inserts ectopically, inferior and medial, and is obstructed with a ureterocele. The lower pole ureter inserts laterally and refluxes.

#152 · Abdominal and Pelvic

What is the ultrasound or urographic appearance of a ureterocele?

Answer

A cobra head sign.

#153 · Abdominal and Pelvic

What distinguishes a collecting system tumour from a blood clot on CT urography?

Answer

Tumour enhances and is fixed to the wall. A clot does not enhance.

#154 · Abdominal and Pelvic

What sequence is key for bladder cancer muscle invasion and what scoring system applies?

Answer

T2-weighted imaging with DWI and DCE, scored using VI-RADS from 1 to 5.

#155 · Abdominal and Pelvic

How does intraperitoneal bladder rupture differ from extraperitoneal on cystography?

Answer

Intraperitoneal shows contrast outlining bowel loops and needs surgery. Extraperitoneal shows flame-shaped perivesical extravasation, usually managed with a catheter.

#156 · Abdominal and Pelvic

What unenhanced HU value diagnoses a lipid-rich adrenal adenoma?

Answer

10 HU or less.

#157 · Abdominal and Pelvic

What are the CT washout thresholds for an adrenal adenoma?

Answer

Absolute washout 60 per cent or more, or relative washout 40 per cent or more, on a 15 minute delayed scan.

#158 · Abdominal and Pelvic

How is absolute percentage washout calculated?

Answer

(Enhanced minus delayed) divided by (enhanced minus unenhanced).

#159 · Abdominal and Pelvic

What does a signal drop on opposed-phase MRI indicate in an adrenal lesion?

Answer

Intracellular lipid, suggesting adenoma. A lipid-poor adenoma may not drop.

#160 · Abdominal and Pelvic

What biochemical test should precede biopsy of a hypervascular, T2-bright adrenal mass?

Answer

Plasma or urinary metanephrines to exclude phaeochromocytoma.

#161 · Abdominal and Pelvic

What imaging feature identifies adrenal myelolipoma?

Answer

Macroscopic fat within the adrenal lesion.

#162 · Abdominal and Pelvic

At what size is an indeterminate adrenal mass considered for resection?

Answer

Above 4 cm.

#163 · Abdominal and Pelvic

How does retroperitoneal fibrosis differ from lymphoma around the aorta?

Answer

Fibrosis hugs the aorta, does not displace it anteriorly and draws ureters medially. Lymphoma lifts the aorta from the spine.

#164 · Abdominal and Pelvic

What does active retroperitoneal fibrosis look like on MRI?

Answer

Bright on T2-weighted imaging with enhancement. Mature fibrosis is dark.

#165 · Abdominal and Pelvic

Which retroperitoneal compartment contains the pancreas and duodenum?

Answer

The anterior pararenal space.

#166 · Abdominal and Pelvic

Which zone is judged mainly on DWI in PI-RADS v2.1, and which on T2?

Answer

The peripheral zone on DWI, the transition zone on T2-weighted imaging.

#167 · Abdominal and Pelvic

What is the role of DCE in PI-RADS v2.1?

Answer

It only upgrades a peripheral zone DWI score of 3 to 4 when positive.

#168 · Abdominal and Pelvic

What size and features define a PI-RADS 5 lesion?

Answer

A lesion 1.5 cm or larger, or with definite extraprostatic extension. It scores 5 on the dominant sequence: marked restriction on DWI in the peripheral zone, or T2 score 5 in the transition zone.

#169 · Abdominal and Pelvic

What T2 appearance scores 2 in the transition zone?

Answer

A circumscribed encapsulated nodule or a mildly heterogeneous nodule.

#170 · Abdominal and Pelvic

Which stage is seminal vesicle invasion in prostate cancer?

Answer

T3b.

#171 · Abdominal and Pelvic

Why can MRI be misleading soon after prostate biopsy?

Answer

Haemorrhage is high on T1 and can mimic or obscure tumour.

#172 · Abdominal and Pelvic

Which imaging is more sensitive than CT for nodal and bone prostate disease?

Answer

PSMA PET-CT.

#173 · Abdominal and Pelvic

What is the imaging test of choice for scrotal disease?

Answer

Ultrasound with colour Doppler.

#174 · Abdominal and Pelvic

What is the typical ultrasound appearance of seminoma?

Answer

A homogeneous hypoechoic, often lobulated, intratesticular mass.

#175 · Abdominal and Pelvic

Where does the right testis drain first?

Answer

To the interaortocaval nodes just below the right renal vein.

#176 · Abdominal and Pelvic

Where does the left testis drain first?

Answer

To the left para-aortic nodes just below the left renal vein.

#177 · Abdominal and Pelvic

What are the Doppler and grey-scale features of testicular torsion?

Answer

Absent or reduced intratesticular flow, an enlarged heterogeneous hypoechoic testis and a whirlpool sign of the cord.

#178 · Abdominal and Pelvic

What is the commonest testicular tumour in men over 60?

Answer

Lymphoma, which can be bilateral.

#179 · Abdominal and Pelvic

Why is a left varicocele more common, and what does a new right varicocele suggest?

Answer

The left testicular vein drains into the left renal vein at a right angle. A new right or non-reducing varicocele prompts a search for a retroperitoneal or renal mass.

#180 · Abdominal and Pelvic

What nodal short-axis threshold is usually used in the abdomen and pelvis?

Answer

More than 10 mm, with round shape, loss of the fatty hilum and necrosis favouring malignancy.

#181 · Musculoskeletal

What defines a Salter-Harris type II injury?

Answer

A fracture through the physis that exits through the metaphysis, leaving a Thurston Holland fragment. It is the commonest type.

#182 · Musculoskeletal

Which Salter-Harris types are intra-articular and need anatomical reduction?

Answer

Types III and IV, because they cross the epiphysis into the joint.

#183 · Musculoskeletal

How is the Weber classification of ankle fractures defined?

Answer

By the level of the fibular fracture relative to the syndesmosis: A below, B at, C above it. Weber C implies syndesmotic disruption and is usually unstable.

#184 · Musculoskeletal

Which proximal injury must be sought when the ankle mortise is widened with a normal distal fibula?

Answer

A Maisonneuve injury, a proximal fibular fracture with deltoid or medial malleolar injury.

#185 · Musculoskeletal

How are Garden stages grouped practically?

Answer

Undisplaced (I and II) versus displaced (III and IV). Displaced fractures carry a high risk of avascular necrosis and are often treated by arthroplasty in older patients.

#186 · Musculoskeletal

What is a Galeazzi injury?

Answer

A distal radial shaft fracture with disruption of the distal radioulnar joint. A Monteggia injury is a proximal ulnar fracture with radial head dislocation.

#187 · Musculoskeletal

What is the best test for an occult scaphoid fracture after normal radiographs?

Answer

MRI, which is the most sensitive. CT is an alternative where MRI is unavailable.

#188 · Musculoskeletal

Why is the proximal pole of the scaphoid prone to avascular necrosis?

Answer

The blood supply enters distally, so a proximal fracture can leave the pole poorly perfused.

#189 · Musculoskeletal

What defines a Salter-Harris type V injury?

Answer

A crush injury to the physis, usually diagnosed in retrospect from growth arrest.

#190 · Musculoskeletal

Which features of a fracture suggest it is pathological?

Answer

Minimal trauma, a lytic lesion or soft tissue mass, or cortical destruction rather than a simple break.

#191 · Musculoskeletal

Which zone of transition and periosteal reaction suggest an aggressive bone lesion?

Answer

A wide, moth-eaten or permeative zone of transition with lamellated, spiculated or interrupted (Codman triangle) periosteal reaction.

#192 · Musculoskeletal

What is the likely diagnosis of an eccentric lytic subarticular lesion in a young adult with a closed physis?

Answer

Giant cell tumour, with a non-sclerotic margin extending to the subarticular surface.

#193 · Musculoskeletal

What are the imaging features of osteoid osteoma?

Answer

A nidus under 2 cm with surrounding sclerosis in a young patient with night pain relieved by NSAIDs. CT shows the nidus best, and radiofrequency ablation treats it.

#194 · Musculoskeletal

What feature of an osteochondroma raises concern for chondrosarcoma?

Answer

A cartilage cap thicker than 1.5 to 2 cm in an adult, or growth after skeletal maturity.

#195 · Musculoskeletal

Which features favour chondrosarcoma over enchondroma in a long bone?

Answer

Pain, endosteal scalloping of more than two thirds of cortical thickness, cortical destruction, a soft tissue mass and size over 5 to 6 cm.

#196 · Musculoskeletal

What is the shepherd's crook deformity and which lesion causes it?

Answer

Bowing of the proximal femur in fibrous dysplasia, a ground-glass lesion with a rind of sclerosis.

#197 · Musculoskeletal

How do a simple bone cyst and an aneurysmal bone cyst differ on imaging?

Answer

A simple cyst is central and lytic, abutting the physis, and may show a fallen fragment sign; an aneurysmal cyst is eccentric, expansile and multiloculated with fluid-fluid levels.

#198 · Musculoskeletal

Where does osteosarcoma typically arise and what is its matrix?

Answer

In the metaphysis around the knee or proximal humerus in adolescents, producing osteoid matrix with a sunburst or Codman triangle reaction.

#199 · Musculoskeletal

Which tumour shows a large soft tissue mass out of proportion to bony destruction in a child?

Answer

Ewing sarcoma, with permeative destruction and lamellated periosteal reaction. It can mimic osteomyelitis.

#200 · Musculoskeletal

Why can a bone scan be normal in myeloma?

Answer

Myeloma produces little osteoblastic activity, so lesions are lytic and punched-out; whole-body low-dose CT or MRI is preferred.

#201 · Musculoskeletal

What is the winking owl sign?

Answer

Loss of a vertebral pedicle on the frontal spine radiograph, from metastatic destruction.

#202 · Musculoskeletal

Why is haematogenous osteomyelitis in children usually metaphyseal?

Answer

Slow vascular flow in the metaphysis allows bacteria to settle.

#203 · Musculoskeletal

When do radiographs become abnormal in acute osteomyelitis and what is the best early test?

Answer

After about 10 to 14 days; MRI is the most sensitive and specific early test.

#204 · Musculoskeletal

Define sequestrum, involucrum and cloaca.

Answer

The sequestrum is necrotic bone separated from viable bone, the involucrum is the new periosteal bone sheath around it, and the cloaca is the channel through which pus drains.

#205 · Musculoskeletal

How does tuberculous spondylitis differ from pyogenic discitis?

Answer

It spares the disc until late, involves the anterior vertebral bodies and produces large paravertebral abscesses and a gibbus.

#206 · Musculoskeletal

What T-score defines osteoporosis on DXA?

Answer

A T-score of minus 2.5 or lower in postmenopausal women and men over 50; osteopenia lies between minus 1 and minus 2.5.

#207 · Musculoskeletal

What are Looser zones and in which condition do they occur?

Answer

Pseudofractures perpendicular to the cortex, often symmetrical, in osteomalacia.

#208 · Musculoskeletal

Describe the three phases of Paget disease.

Answer

A lytic phase (flame-shaped lucency, osteoporosis circumscripta), a mixed phase, and a sclerotic phase with cortical thickening, coarse trabeculae and bone expansion.

#209 · Musculoskeletal

How do osteoarthritis and rheumatoid arthritis differ in bone density and joint distribution?

Answer

Osteoarthritis preserves density and affects the DIP, PIP and first CMC joints; rheumatoid causes periarticular osteopenia with marginal erosions of the MCP, PIP and wrists, sparing the DIPs.

#210 · Musculoskeletal

What are the radiographic features of gout?

Answer

Punched-out erosions with overhanging edges away from the joint margin, preserved joint space until late, soft tissue tophi and normal density.

#211 · Musculoskeletal

What is the earliest imaging finding in ankylosing spondylitis and where?

Answer

Bilateral symmetrical sacroiliitis with iliac-side erosions and sclerosis; MRI shows marrow oedema before radiographs change.

#212 · Musculoskeletal

What are the MRI signs of femoral head avascular necrosis?

Answer

A serpentine low T1 band and the double line sign on T2; later radiographs show sclerosis, the crescent sign and collapse.

#213 · Musculoskeletal

How is slipped capital femoral epiphysis recognised on an AP pelvis?

Answer

The Klein line along the superior femoral neck fails to intersect the epiphysis; the lateral view is more sensitive.

#214 · Musculoskeletal

Which findings in a child have high specificity for non-accidental injury?

Answer

Classic metaphyseal lesions (corner or bucket handle), posterior rib fractures and scapular fractures. Fractures of differing ages are only moderately specific.

#215 · Musculoskeletal

How is developmental dysplasia of the hip screened in infants?

Answer

Ultrasound before ossification of the femoral head (about 4 to 6 months), assessing the alpha angle and femoral head coverage.

#216 · Musculoskeletal

What MRI criterion defines a meniscal tear?

Answer

Intermeniscal signal reaching an articular surface, on at least two consecutive slices or in two planes. Signal confined within the meniscus is intrasubstance degeneration.

#217 · Musculoskeletal

Which signs indicate a bucket handle meniscal tear?

Answer

The double PCL sign from the flipped inner fragment in the intercondylar notch, and a truncated or absent bow tie on sagittal images of the body.

#218 · Musculoskeletal

Why does a medial meniscus root radial tear matter biomechanically?

Answer

It disrupts hoop tension, so the meniscus behaves as if resected, with extrusion of the body and often subchondral insufficiency change.

#219 · Musculoskeletal

Which bone bruise pattern accompanies an acute ACL tear?

Answer

Pivot shift contusions in the lateral femoral condyle and the posterolateral tibial plateau, often with anterior tibial translation.

#220 · Musculoskeletal

Which small avulsion fracture is strongly associated with an ACL tear?

Answer

The Segond fracture, an avulsion from the lateral tibial plateau reflecting lateral capsular injury.

#221 · Musculoskeletal

How is an MCL injury graded on MRI?

Answer

Grade 1 is periligamentous oedema, grade 2 is partial fibre disruption, grade 3 is complete discontinuity.

#222 · Musculoskeletal

What is the kissing contusion pattern of transient patellar dislocation?

Answer

Oedema in the medial patellar facet and the anterolateral femoral condyle, with a torn medial patellofemoral ligament and an effusion.

#223 · Musculoskeletal

Which condyle region is typical for osteochondritis dissecans of the knee?

Answer

The lateral aspect of the medial femoral condyle, in adolescents.

#224 · Musculoskeletal

What MRI features make an osteochondritis dissecans fragment unstable?

Answer

A T2 bright fluid line beneath the fragment, a cyst beneath it, a defect in the overlying cartilage and fluid tracking through it.

#225 · Musculoskeletal

Which rotator cuff muscle finding predicts poor repair outcome?

Answer

High grade Goutallier fatty infiltration on sagittal images, with a tangent sign and tendon retraction.

#226 · Musculoskeletal

Which anatomical factors predispose to subacromial impingement?

Answer

A type III hooked acromion, an os acromiale and a degenerate acromioclavicular joint with inferior osteophytes.

#227 · Musculoskeletal

How does calcific tendinitis appear on imaging?

Answer

Dense on radiographs, low signal on all MRI sequences, usually in the supraspinatus 1 to 2 cm from the insertion, with oedema in the resorptive phase.

#228 · Musculoskeletal

What injuries result from anterior shoulder dislocation?

Answer

A Bankart tear of the anteroinferior labrum and a Hill-Sachs impaction fracture of the posterolateral humeral head.

#229 · Musculoskeletal

What distinguishes an ALPSA lesion from a Perthes lesion?

Answer

In ALPSA the labrum and periosteum are stripped and displaced medially along the scapular neck. In a Perthes lesion the stripped periosteal sleeve is not displaced.

#230 · Musculoskeletal

Which injury pattern is classic for posterior shoulder dislocation after seizure or electric shock?

Answer

A reverse Bankart lesion with a reverse Hill-Sachs defect in the anteromedial humeral head.

#231 · Musculoskeletal

How is cam morphology quantified and what is it?

Answer

An aspherical femoral head neck junction measured by an alpha angle above about 55 degrees on axial oblique images.

#232 · Musculoskeletal

Which Modic type is the commonest and what does it represent?

Answer

Type 2 is the commonest. It is high signal on both T1 and T2 and represents fatty replacement of the endplate marrow.

#233 · Musculoskeletal

Which ankle ligament is most often injured in an inversion sprain?

Answer

The anterior talofibular ligament, then the calcaneofibular ligament.

#234 · Musculoskeletal

Which ligament is involved in a high ankle sprain?

Answer

The anterior inferior tibiofibular ligament of the syndesmosis, typically after external rotation.

#235 · Musculoskeletal

Where are talar dome osteochondral lesions found?

Answer

Posteromedially after inversion with plantar flexion, and anterolaterally after inversion with dorsiflexion.

#236 · Musculoskeletal

What does a scapholunate gap above 3 mm on a radiograph indicate?

Answer

The Terry Thomas sign of scapholunate ligament disruption, which can progress to scapholunate advanced collapse.

#237 · Musculoskeletal

What is the zoning phenomenon of myositis ossificans?

Answer

Peripheral mature ossification with an immature centre on CT, which distinguishes it from sarcoma. Early lesions can mimic a tumour.

#238 · Musculoskeletal

How are MRI muscle strains graded?

Answer

Grade 1 is oedema without architectural distortion, grade 2 is a partial tear with haematoma, grade 3 is a complete rupture with a gap.

#239 · Musculoskeletal

Which MRI features favour atypical lipomatous tumour over lipoma?

Answer

Size over 10 cm, deep location, septa thicker than 2 mm, nodularity and non-fatty components, with MDM2 amplification confirming it.

#240 · Musculoskeletal

What are the MRI signs of a peripheral nerve sheath tumour?

Answer

The target sign, the split fat sign and the fascicular sign, along the course of a nerve.

#241 · Musculoskeletal

Why does tenosynovial giant cell tumour bloom on gradient echo?

Answer

Haemosiderin deposition causes low T1 and T2 signal with susceptibility blooming and synovial enhancement.

#242 · Musculoskeletal

What are the Modic endplate types on T1 and T2?

Answer

Type 1 is low T1 and high T2 (oedema), type 2 is high on both (fat), type 3 is low on both (sclerosis).

#243 · Musculoskeletal

How are disc herniations classified by shape?

Answer

Protrusion has a base wider than any other dimension of the herniated material, extrusion has a dimension exceeding the base, and sequestration is a free fragment.

#244 · Musculoskeletal

Which nerve roots are compressed by a posterolateral L4/5 disc extrusion and a far lateral L4/5 herniation?

Answer

A posterolateral extrusion compresses the traversing L5 root in the lateral recess. A far lateral herniation compresses the exiting L4 root.

#245 · Musculoskeletal

How is lumbar spondylolisthesis graded?

Answer

By Meyerding in quarters of vertebral body width, from grade I up to 25 per cent through grade V, spondyloptosis.

#246 · Musculoskeletal

What are the types of odontoid fracture, and which is least likely to unite?

Answer

Anderson and D'Alonzo: type I tip avulsion, type II through the base, type III into the C2 body. Type II has the highest non-union risk.

#247 · Musculoskeletal

What distinguishes pyogenic spondylodiscitis from Modic type 1 change?

Answer

Infection shows a T2 bright disc with loss of the intranuclear cleft, endplate erosion and collections. Modic 1 has a dark disc and intact endplate cortex.

#248 · Musculoskeletal

How does tuberculous spondylitis differ from pyogenic infection?

Answer

Subligamentous spread over several vertebrae, relative disc preservation, large calcified paraspinal abscess, thoracic predilection and gibbus deformity.

#249 · Musculoskeletal

How does DISH differ from ankylosing spondylitis?

Answer

DISH has flowing ossification over four or more contiguous vertebrae with preserved discs and normal sacroiliac joints, typically right-sided in the thoracic spine.

#250 · Musculoskeletal

What MRI finding qualifies as active sacroiliitis in axial spondyloarthritis?

Answer

Subchondral bone marrow oedema on STIR, on at least two slices if a single lesion or one slice if multiple lesions. Erosions, sclerosis, fat metaplasia and ankylosis are structural lesions.

#251 · Neuroradiology and Head and Neck

What does ASPECTS measure and what is the maximum score?

Answer

The extent of early ischaemic change in the MCA territory on non contrast CT, out of 10 with one point deducted per affected region. A score of 6 or more clearly favours thrombectomy; scores of 3 to 5 may still benefit in recent large core trials.

#252 · Neuroradiology and Head and Neck

What CT finding suggests fresh thrombus in the MCA?

Answer

A hyperdense MCA sign. In cross section within the Sylvian fissure it is called the dot sign.

#253 · Neuroradiology and Head and Neck

How is the penumbra defined on CT perfusion?

Answer

As tissue with delayed Tmax (commonly above 6 seconds) beyond the core, with relatively preserved blood volume. The core is where cerebral blood volume or flow is reduced.

#254 · Neuroradiology and Head and Neck

What does DWI FLAIR mismatch indicate?

Answer

A DWI lesion without matching FLAIR hyperintensity suggests onset within about 4.5 hours, useful in wake up stroke.

#255 · Neuroradiology and Head and Neck

Which imaging features suggest cerebral venous sinus thrombosis?

Answer

A dense sinus on non contrast CT, the empty delta sign on contrast CT, and venous infarcts that cross arterial territories, often haemorrhagic. MR or CT venography confirms it.

#256 · Neuroradiology and Head and Neck

Where do hypertensive haemorrhages typically occur?

Answer

In the putamen, thalamus, pons and cerebellum.

#257 · Neuroradiology and Head and Neck

Which pattern suggests cerebral amyloid angiopathy?

Answer

Lobar haemorrhage in an elderly patient with multiple peripheral cortical microbleeds on SWI.

#258 · Neuroradiology and Head and Neck

What does the spot sign on CTA predict?

Answer

Haematoma expansion, because contrast within the haematoma marks active bleeding.

#259 · Neuroradiology and Head and Neck

What is the MRI appearance of early subacute haematoma?

Answer

Intracellular methaemoglobin: bright on T1 and dark on T2.

#260 · Neuroradiology and Head and Neck

Which MRI sequences show chronic haematoma as dark?

Answer

T2 and SWI, from haemosiderin that persists as a rim for years.

#261 · Neuroradiology and Head and Neck

How sensitive is early CT for SAH?

Answer

Close to 100 percent within 6 hours of onset on a modern scanner, then falling with time, so lumbar puncture may be needed later.

#262 · Neuroradiology and Head and Neck

Which aneurysm does blood in the interhemispheric fissure suggest?

Answer

An anterior communicating artery aneurysm, the most common site.

#263 · Neuroradiology and Head and Neck

What is the typical pattern of perimesencephalic non aneurysmal SAH?

Answer

Blood centred anterior to the midbrain and pons with little Sylvian extension, a benign course, and a negative CTA.

#264 · Neuroradiology and Head and Neck

When does vasospasm typically occur after SAH?

Answer

From about day 4, peaking between days 7 and 10.

#265 · Neuroradiology and Head and Neck

Which conditions are associated with intracranial berry aneurysms?

Answer

Autosomal dominant polycystic kidney disease, vascular Ehlers Danlos syndrome, coarctation of the aorta and fibromuscular dysplasia.

#266 · Neuroradiology and Head and Neck

What three features determine the Spetzler Martin grade of an AVM?

Answer

Nidus size, eloquent location, and deep venous drainage.

#267 · Neuroradiology and Head and Neck

What is the classic MRI appearance of a cavernous malformation?

Answer

Popcorn lesion with a complete hemosiderin rim dark on T2 and SWI, angiographically occult with no flow voids.

#268 · Neuroradiology and Head and Neck

What is a developmental venous anomaly and how does it appear?

Answer

A benign variant with a caput medusae of small medullary veins converging on a large draining vein. It drains normal brain and should be preserved.

#269 · Neuroradiology and Head and Neck

Which imaging features favour glioblastoma over metastasis?

Answer

A solitary thick irregular enhancing ring with central necrosis, possibly crossing the corpus callosum, rather than multiple lesions at the grey white junction.

#270 · Neuroradiology and Head and Neck

How does primary CNS lymphoma appear?

Answer

Periventricular, hyperdense on CT, homogeneously enhancing with marked restricted diffusion, and steroid sensitive.

#271 · Neuroradiology and Head and Neck

Which sign favours toxoplasmosis over lymphoma in HIV?

Answer

An eccentric target sign with low perfusion. Lymphoma tends to be single with higher perfusion and subependymal spread.

#272 · Neuroradiology and Head and Neck

Which features indicate a meningioma?

Answer

Broad dural base, dural tail, CSF cleft, hyperostosis, calcification and avid homogeneous enhancement.

#273 · Neuroradiology and Head and Neck

How does medulloblastoma appear in a child?

Answer

A midline vermian fourth ventricle mass, hyperdense on CT with restricted diffusion and drop metastases.

#274 · Neuroradiology and Head and Neck

Which paediatric tumour is cystic with an enhancing mural nodule in the cerebellum?

Answer

Pilocytic astrocytoma.

#275 · Neuroradiology and Head and Neck

What is the classic lesion orientation in multiple sclerosis?

Answer

Ovoid T2 and FLAIR bright lesions perpendicular to the lateral ventricles, called Dawson fingers.

#276 · Neuroradiology and Head and Neck

What does dissemination in time mean in the 2017 McDonald criteria?

Answer

Simultaneous enhancing and non enhancing lesions, or a new lesion on follow up imaging (McDonald 2017, revised 2024).

#277 · Neuroradiology and Head and Neck

Which MRI finding distinguishes NMOSD from MS in the cord?

Answer

Longitudinally extensive transverse myelitis spanning three or more vertebral segments, with AQP4 antibodies.

#278 · Neuroradiology and Head and Neck

What are the features of PML on MRI?

Answer

Subcortical white matter and U fibre lesions with no mass effect and usually no enhancement, in immunosuppressed patients.

#279 · Neuroradiology and Head and Neck

Which areas does herpes simplex encephalitis involve?

Answer

Medial temporal lobes, insula and inferior frontal lobes, usually asymmetric, sparing the basal ganglia.

#280 · Neuroradiology and Head and Neck

Which sign distinguishes an abscess from a necrotic tumour on SWI?

Answer

The dual rim sign supports abscess, which also shows central restricted diffusion and a thinner wall on the ventricular side.

#281 · Neuroradiology and Head and Neck

What does the MRI show in Creutzfeldt Jakob disease?

Answer

Cortical ribboning and basal ganglia restricted diffusion on DWI, with the pulvinar sign in variant CJD.

#282 · Neuroradiology and Head and Neck

What pattern of atrophy suggests Alzheimer disease?

Answer

Medial temporal and hippocampal atrophy followed by posterior parietal atrophy.

#283 · Neuroradiology and Head and Neck

Which sign is classic for progressive supranuclear palsy?

Answer

Midbrain atrophy giving the hummingbird or penguin sign on sagittal images.

#284 · Neuroradiology and Head and Neck

What does DAT SPECT show in Parkinson disease?

Answer

Reduced striatal uptake, whereas essential tremor and drug induced parkinsonism are normal.

#285 · Neuroradiology and Head and Neck

What imaging signs define normal pressure hydrocephalus?

Answer

Ventriculomegaly out of proportion to sulci, callosal angle below 90 degrees, DESH pattern and increased aqueductal flow void.

#286 · Neuroradiology and Head and Neck

What is the typical location and CT density of a colloid cyst?

Answer

Anterosuperior third ventricle at the foramen of Monro, usually hyperdense, causing obstructive hydrocephalus.

#287 · Neuroradiology and Head and Neck

What are the CT features of an extradural haematoma?

Answer

Biconvex, does not cross sutures, may cross dural reflections, from the middle meningeal artery with an overlying temporoparietal fracture.

#288 · Neuroradiology and Head and Neck

How does a subdural haematoma differ from an extradural haematoma?

Answer

Crescentic, crosses sutures, limited by dural reflections such as the falx, and varies in density with age.

#289 · Neuroradiology and Head and Neck

Where does diffuse axonal injury occur?

Answer

At the grey white junction, corpus callosum especially the splenium, and dorsolateral brainstem, best seen on SWI and DWI.

#290 · Neuroradiology and Head and Neck

How is a pituitary microadenoma defined and how does it enhance?

Answer

Under 10 mm, and enhances less avidly than the normal gland on dynamic contrast imaging.

#291 · Neuroradiology and Head and Neck

What does the Knosp classification assess?

Answer

Cavernous sinus invasion by a pituitary macroadenoma, with grade 4 meaning encasement of the intracavernous internal carotid artery.

#292 · Neuroradiology and Head and Neck

How does a craniopharyngioma appear?

Answer

A suprasellar mass with calcification and cystic components, bimodal in children and adults.

#293 · Neuroradiology and Head and Neck

What defines Chiari I malformation?

Answer

Cerebellar tonsillar descent of at least 5 mm below the foramen magnum.

#294 · Neuroradiology and Head and Neck

How do epidermoid and arachnoid cysts differ on DWI?

Answer

Epidermoid restricts diffusion and does not fully suppress on FLAIR, whereas arachnoid cyst follows CSF with no restriction.

#295 · Neuroradiology and Head and Neck

What is the ice cream cone sign?

Answer

A vestibular schwannoma with an intracanalicular stalk and cerebellopontine angle component, widening the internal auditory canal.

#296 · Neuroradiology and Head and Neck

Which MRI feature separates cholesteatoma from granulation tissue in the middle ear?

Answer

Cholesteatoma restricts on non-echo-planar diffusion weighted imaging and does not enhance centrally. Granulation tissue enhances and does not restrict.

#297 · Neuroradiology and Head and Neck

Where does acquired cholesteatoma classically arise, and which ossicular part erodes first?

Answer

It arises in Prussak space (lateral epitympanic recess) and erodes the scutum. The long process of the incus is the ossicular part eroded first.

#298 · Neuroradiology and Head and Neck

What is the signal pattern of a petrous apex cholesterol granuloma?

Answer

It is expansile and smooth, hyperintense on both T1 and T2. A petrous apex effusion or mucocele is T2 bright but T1 dark.

#299 · Neuroradiology and Head and Neck

Where is the lucent focus of fenestral otosclerosis?

Answer

Just anterior to the oval window (fissula ante fenestram), causing conductive hearing loss. Retrofenestral disease around the cochlea causes sensorineural loss.

#300 · Neuroradiology and Head and Neck

How does a glomus jugulare differ from a glomus tympanicum on CT?

Answer

Glomus jugulare expands the jugular foramen and erodes the jugular spine with a permeative margin. A glomus tympanicum sits on the cochlear promontory with the jugular plate intact.

#301 · Neuroradiology and Head and Neck

What size defines a large vestibular aqueduct?

Answer

More than 1.5 mm at the midpoint. It is associated with incomplete partition type II (Mondini) and sensorineural hearing loss.

#302 · Neuroradiology and Head and Neck

Which cerebellopontine angle mass enlarges the internal auditory canal and forms an ice cream cone shape?

Answer

Vestibular schwannoma. A meningioma is broad based with a dural tail and usually does not enlarge the canal.

#303 · Neuroradiology and Head and Neck

Why is an Onodi cell reported before sinus surgery?

Answer

It is a posterior ethmoid cell extending lateral or superior to the sphenoid sinus, placing the optic nerve at risk of injury.

#304 · Neuroradiology and Head and Neck

What is the Chandler classification used for?

Answer

Orbital complications of sinusitis: preseptal cellulitis, orbital cellulitis, subperiosteal abscess, orbital abscess and cavernous sinus thrombosis.

#305 · Neuroradiology and Head and Neck

What is the black turbinate sign and what does it suggest?

Answer

Loss of mucosal enhancement of a turbinate, indicating acute invasive fungal sinusitis in an immunocompromised or diabetic patient. It is an emergency.

#306 · Neuroradiology and Head and Neck

How do allergic fungal sinusitis and mycetoma differ?

Answer

Allergic fungal sinusitis expands several sinuses with central hyperattenuating T2 dark material. Mycetoma is usually a single maxillary sinus with a central dense or calcific focus.

#307 · Neuroradiology and Head and Neck

What are the imaging features of inverted papilloma?

Answer

Arises from the lateral nasal wall near the middle meatus, convoluted cerebriform pattern on contrast MRI, hyperostosis at the attachment. Loss of the pattern suggests squamous cell transformation.

#308 · Neuroradiology and Head and Neck

Which sinonasal tumour affects adolescent males, centres on the sphenopalatine foramen and bows the maxillary sinus posterior wall forward?

Answer

Juvenile nasopharyngeal angiofibroma, with prominent flow voids and pterygopalatine fossa widening.

#309 · Neuroradiology and Head and Neck

Where does olfactory neuroblastoma sit and what is its typical shape?

Answer

At the cribriform plate, dumbbell shaped through the skull base, often with peritumoral cysts at the intracranial margin.

#310 · Neuroradiology and Head and Neck

What shape does retinal detachment make, and how does choroidal detachment differ?

Answer

Retinal detachment is V shaped converging at the optic disc. Choroidal detachment is lentiform and does not reach the disc.

#311 · Neuroradiology and Head and Neck

What is the tram track sign in the orbit?

Answer

Enhancing optic nerve sheath around a non-enhancing nerve, typical of optic nerve sheath meningioma.

#312 · Neuroradiology and Head and Neck

How do thyroid eye disease and idiopathic orbital inflammation differ in muscle involvement?

Answer

Thyroid eye disease enlarges the muscle bellies and spares the tendons (inferior and medial recti commonest). Idiopathic orbital inflammation involves the tendinous insertions and responds rapidly to steroids.

#313 · Neuroradiology and Head and Neck

In which direction do masses of each deep neck space displace the parapharyngeal fat?

Answer

Pharyngeal mucosal: laterally. Masticator: posteriorly. Parotid: medially. Carotid: anteriorly.

#314 · Neuroradiology and Head and Neck

What is the lyre sign?

Answer

Splaying of the internal and external carotid arteries by a carotid body paraganglioma, which is avidly enhancing with flow voids.

#315 · Neuroradiology and Head and Neck

How does a retropharyngeal abscess differ from an effusion, and where can it spread?

Answer

An abscess shows a rim enhancing collection, an effusion does not. Infection can pass through the danger space to the posterior mediastinum.

#316 · Neuroradiology and Head and Neck

Which parotid tumour is T2 bright and the commonest?

Answer

Pleomorphic adenoma. Warthin tumour is in older male smokers, often bilateral in the parotid tail, and takes up technetium pertechnetate.

#317 · Neuroradiology and Head and Neck

Which salivary malignancy is commonest, and which spreads along nerves?

Answer

Mucoepidermoid carcinoma is commonest. Adenoid cystic carcinoma characteristically spreads along nerves.

#318 · Neuroradiology and Head and Neck

How is an ACR TI-RADS nodule scored for composition and echogenicity?

Answer

Composition: cystic or spongiform 0, mixed 1, solid 2. Echogenicity: anechoic 0, iso or hyperechoic 1, hypoechoic 2, very hypoechoic 3.

#319 · Neuroradiology and Head and Neck

Which TI-RADS features score 3 points?

Answer

Very hypoechoic echogenicity, taller than wide shape, extrathyroidal extension, and punctate echogenic foci.

#320 · Neuroradiology and Head and Neck

Which total scores correspond to TR3, TR4 and TR5?

Answer

TR3 is 3 points, TR4 is 4 to 6 points, TR5 is 7 or more. TR1 is 0 and TR2 is 2.

#321 · Neuroradiology and Head and Neck

What are the FNA and follow up size thresholds for TR3, TR4 and TR5?

Answer

TR3: FNA 2.5 cm, follow up 1.5 cm. TR4: FNA 1.5 cm, follow up 1 cm. TR5: FNA 1 cm, follow up 0.5 cm.

#322 · Neuroradiology and Head and Neck

Which laryngeal findings define T3 and T4a disease?

Answer

T3: paraglottic space invasion with cord fixation or inner cortex thyroid cartilage erosion. T4a: through the outer cortex or into neck soft tissue, thyroid or oesophagus.

#323 · Neuroradiology and Head and Neck

What nodal appearance is typical of HPV related oropharyngeal carcinoma?

Answer

Large cystic nodes from a small tonsil or base of tongue primary.

#324 · Neuroradiology and Head and Neck

Where is the primary in nasopharyngeal carcinoma and what is its association?

Answer

Near the fossa of Rosenmuller. It is associated with EBV and spreads to the skull base and retropharyngeal nodes.

#325 · Neuroradiology and Head and Neck

What are the boundaries of nodal levels II, III and IV?

Answer

II: skull base to lower hyoid. III: hyoid to cricoid. IV: cricoid to clavicle.

#326 · Neuroradiology and Head and Neck

Which cranial nerves lie in the cavernous sinus wall versus within the sinus?

Answer

III, IV, V1 and V2 lie in the lateral wall. VI and the internal carotid artery lie within the sinus.

#327 · Neuroradiology and Head and Neck

What imaging signs indicate perineural tumour spread?

Answer

Nerve enlargement and enhancement, a widened or eroded foramen, loss of fat in the pterygopalatine fossa, and later denervation change in the supplied muscle.

#328 · Neuroradiology and Head and Neck

Which cord lesion pattern suggests multiple sclerosis rather than NMOSD or transverse myelitis?

Answer

Short segment (under two vertebral bodies), peripheral dorsal or lateral, under half the cord cross section. NMOSD is three or more segments.

#329 · Neuroradiology and Head and Neck

What does a long central T2 hyperintense cord with dorsal flow voids in an older man suggest?

Answer

A spinal dural arteriovenous fistula, confirmed and located by spinal angiography.

#330 · Neuroradiology and Head and Neck

Which intramedullary tumour shows a cyst with enhancing mural nodule?

Answer

Haemangioblastoma, suggesting von Hippel Lindau disease.

#331 · Neuroradiology and Head and Neck

How do spinal ependymoma and astrocytoma differ in position?

Answer

Ependymoma is central, often with a haemosiderin cap, in adults. Astrocytoma is eccentric and infiltrative, commoner in children.

#332 · Neuroradiology and Head and Neck

Which intradural extramedullary tumour is thoracic, in women, with a dural tail?

Answer

Meningioma. Nerve sheath tumours are dumbbell shaped and widen the exit foramen.

#333 · Neuroradiology and Head and Neck

What combined C1 lateral mass overhang suggests transverse ligament rupture in a Jefferson fracture?

Answer

More than 7 mm on the open mouth view.

#334 · Neuroradiology and Head and Neck

Which odontoid fracture type has the highest nonunion rate?

Answer

Type II, at the junction of the dens and the vertebral body.

#335 · Neuroradiology and Head and Neck

How do flexion and extension teardrop fractures differ?

Answer

Flexion teardrop is unstable, with posterior body displacement and cord injury. Extension teardrop is a small avulsion with anterior longitudinal ligament injury and is usually stable.

#336 · Neuroradiology and Head and Neck

What anterolisthesis suggests unilateral versus bilateral facet dislocation?

Answer

Unilateral: under 25 percent with a bow tie sign. Bilateral: over 50 percent.

#337 · Neuroradiology and Head and Neck

Which findings in Chiari II malformation accompany myelomeningocele?

Answer

Small posterior fossa, beaked tectum, tonsillar and vermian herniation and hydrocephalus.

#338 · Neuroradiology and Head and Neck

What conus level and filum thickness suggest a tethered cord beyond three months of age?

Answer

Conus below L2 and a filum thicker than 2 mm.

#339 · Neuroradiology and Head and Neck

How is carotid stenosis measured by NASCET, and which duplex velocity suggests 70 percent or more?

Answer

One minus narrowest residual lumen over normal distal ICA lumen. Peak systolic velocity over 230 cm/s.

#340 · Neuroradiology and Head and Neck

How does cervical ICA dissection differ from atherosclerosis in location?

Answer

Dissection starts about 2 cm beyond the bulb and spares it, with a crescent T1 hyperintense mural haematoma. Atherosclerosis centres on the bulb.

#341 · Breast

What are the three BI-RADS mass shapes and which is most suspicious?

Answer

Oval, round and irregular. Irregular shape is the most suspicious.

#342 · Breast

Which mass margin is most strongly associated with invasive carcinoma?

Answer

Spiculated margin, lines radiating from the mass.

#343 · Breast

What defines a circumscribed margin in BI-RADS?

Answer

The margin is sharply demarcated with an abrupt transition from adjacent tissue, with no microlobulated, indistinct or spiculated portion.

#344 · Breast

How does a focal asymmetry differ from an asymmetry?

Answer

An asymmetry is seen on one projection only. A focal asymmetry is seen on two projections but lacks the convexity and conspicuity of a mass.

#345 · Breast

What makes an asymmetry a developing asymmetry, and why does it matter?

Answer

It is new, larger or more conspicuous than on prior studies. It is suspicious.

#346 · Breast

Name the typically benign calcification types.

Answer

Skin, vascular, coarse or popcorn, large rod like, round, rim or eggshell, dystrophic, milk of calcium and suture.

#347 · Breast

How does milk of calcium appear on CC and ML views?

Answer

Round smudges on the craniocaudal view and crescent or meniscus shaped curves on the true lateral view, because it layers dependently in microcysts.

#348 · Breast

List suspicious calcification morphologies from least to most suspicious.

Answer

Amorphous, coarse heterogeneous, fine pleomorphic, fine linear or fine linear branching.

#349 · Breast

What does fine linear branching calcification imply?

Answer

Calcification filling a duct involved by tumour, classically high grade DCIS with comedonecrosis.

#350 · Breast

Which calcification distributions raise the probability of malignancy most?

Answer

Linear and segmental distributions, which suggest deposits in a duct and its branches.

#351 · Breast

What BI-RADS category usually follows grouped amorphous calcifications?

Answer

Category 4B.

#352 · Breast

What is the malignancy likelihood range for each BI-RADS category 3, 4A, 4B, 4C and 5?

Answer

3: 2 percent or less. 4A: more than 2 to 10 percent. 4B: more than 10 to 50 percent. 4C: more than 50 to less than 95 percent. 5: at least 95 percent.

#353 · Breast

What does BI-RADS 6 mean?

Answer

Known biopsy proven malignancy before definitive therapy.

#354 · Breast

Which findings may be assigned BI-RADS 3 after full diagnostic work up?

Answer

A non calcified circumscribed solid mass on ultrasound, a stable or decreasing focal asymmetry and solitary grouped punctate calcifications.

#355 · Breast

What is the follow up schedule for a BI-RADS 3 lesion?

Answer

6 months, then 12 and 24 months, downgrading to category 2 after two years of stability.

#356 · Breast

Why does tomosynthesis improve breast screening?

Answer

It reduces tissue overlap, lowers recall rates and increases detection of invasive cancer, especially in dense breasts.

#357 · Breast

How are BI-RADS breast density categories A to D defined?

Answer

A almost entirely fatty, B scattered fibroglandular, C heterogeneously dense, D extremely dense.

#358 · Breast

Which age group does BreastScreen Australia actively invite, and how often?

Answer

Women aged 50 to 74, every two years.

#359 · Breast

Who qualifies for annual screening MRI?

Answer

High risk women such as BRCA1 or BRCA2 carriers, untested first degree relatives of carriers and those with chest irradiation at a young age.

#360 · Breast

What are the features of a simple breast cyst on ultrasound?

Answer

Anechoic, thin imperceptible wall, circumscribed margin and posterior acoustic enhancement. It is category 2.

#361 · Breast

Which ultrasound orientation is suspicious for a solid mass?

Answer

Not parallel to the skin, taller than wide.

#362 · Breast

What features favour a fibroadenoma on ultrasound?

Answer

Oval, parallel, circumscribed, mildly hypoechoic solid mass, typically in a young woman.

#363 · Breast

What cortical thickness is normal for an axillary node on ultrasound?

Answer

Less than 3 mm with a fatty hilum.

#364 · Breast

What non mass enhancement pattern is classic for DCIS on MRI?

Answer

Segmental, clumped enhancement following a duct.

#365 · Breast

Which MRI finding takes priority when morphology and kinetics conflict?

Answer

Morphology outweighs kinetic curve behaviour.

#366 · Breast

Name three indications for breast MRI.

Answer

Staging in lobular cancer or dense breasts, response to neoadjuvant chemotherapy, high risk screening, equivocal findings, implant integrity, or occult primary with axillary metastasis.

#367 · Breast

When should a marker clip be placed at biopsy?

Answer

Always, particularly if neoadjuvant chemotherapy is planned, as the lesion may resolve completely.

#368 · Breast

What do the Australian B categories B1 to B5 mean?

Answer

B1 normal or non diagnostic, B2 benign, B3 uncertain malignant potential, B4 suspicious, B5 malignant (5a in situ, 5b invasive).

#369 · Breast

What is the action for a benign core result from a BI-RADS 5 lesion?

Answer

It is discordant: repeat biopsy or excise.

#370 · Breast

Which B3 lesion is usually excised because of its upgrade risk to DCIS or invasive cancer?

Answer

Atypical ductal hyperplasia.

#371 · Breast

What is the most reliable MRI sign of intracapsular silicone implant rupture?

Answer

The linguine sign, curvilinear low signal lines within the silicone.

#372 · Breast

How does extracapsular silicone rupture appear on ultrasound?

Answer

A snowstorm pattern of echogenic noise with posterior shadowing.

#373 · Breast

How does BIA-ALCL typically present, and how is it confirmed?

Answer

A late periprosthetic fluid collection, usually over a year after surgery; aspirate cytology with CD30 immunostaining.

#374 · Breast

What is the typical mammographic appearance of gynaecomastia versus male breast cancer?

Answer

Gynaecomastia is a subareolar flame shaped density radiating from the nipple. Cancer is an eccentric, high density, often spiculated mass.

#375 · Breast

How are axillary nodal levels defined?

Answer

Relative to pectoralis minor: level I lateral, level II behind, level III medial to its medial border.

#376 · Obstetrics and Gynaecology

What is the first reliable ultrasound sign of an intrauterine pregnancy?

Answer

A visible yolk sac within the gestational sac, usually from about 5.5 weeks. A sac without a yolk sac may be a pseudosac of an ectopic pregnancy.

#377 · Obstetrics and Gynaecology

How does a tubal ectopic differ from a corpus luteum on transvaginal ultrasound?

Answer

A corpus luteum lies within the ovary and moves with it. A tubal ectopic is an adnexal mass with a thick echogenic ring that moves independently of the ovary.

#378 · Obstetrics and Gynaecology

What is the most common site of tubal ectopic pregnancy?

Answer

The ampulla of the fallopian tube.

#379 · Obstetrics and Gynaecology

Which findings suggest an interstitial ectopic pregnancy?

Answer

An eccentric sac in the fundus with a thin myometrial mantle, commonly under 5 mm, and an interstitial line sign pointing to the sac.

#380 · Obstetrics and Gynaecology

Where does a caesarean scar ectopic implant and why does it matter?

Answer

In the scar niche of the lower anterior uterine segment, with thin myometrium toward the bladder. It risks haemorrhage and later placenta accreta spectrum.

#381 · Obstetrics and Gynaecology

What crown rump length and heartbeat finding is diagnostic of failed early pregnancy?

Answer

A crown rump length of 7 mm or more with no cardiac activity.

#382 · Obstetrics and Gynaecology

What mean sac diameter with no embryo is diagnostic of failed pregnancy?

Answer

A mean sac diameter of 25 mm or more with no embryo.

#383 · Obstetrics and Gynaecology

Which early pregnancy findings are suspicious but not diagnostic of failure?

Answer

Crown rump length under 7 mm without a heartbeat, mean sac diameter 16 to 24 mm with no embryo, and an enlarged yolk sac over 7 mm. A repeat scan after 7 to 10 days is advised.

#384 · Obstetrics and Gynaecology

What does a markedly vascular myometrial lesion after curettage suggest?

Answer

An acquired arteriovenous malformation, with low resistance high velocity flow. Biopsy or curettage risks severe bleeding.

#385 · Obstetrics and Gynaecology

Up to what crown rump length is it used for dating?

Answer

About 84 mm, roughly 14 weeks. After that head circumference is used.

#386 · Obstetrics and Gynaecology

When is the nuchal translucency scan performed and what threshold is used?

Answer

From 11 weeks 0 days to 13 weeks 6 days, with crown rump length 45 to 84 mm. A thickness of 3.5 mm or more prompts further testing.

#387 · Obstetrics and Gynaecology

How is fetal growth restriction distinguished from a small for gestational age fetus?

Answer

Small for gestational age is an estimated weight under the 10th centile. Growth restriction needs a small abdominal circumference, weight under the 3rd centile or abnormal Doppler indices.

#388 · Obstetrics and Gynaecology

What umbilical artery Doppler sequence indicates worsening placental insufficiency?

Answer

Raised resistance, then absent end diastolic flow, then reversed end diastolic flow.

#389 · Obstetrics and Gynaecology

Which middle cerebral artery finding suggests fetal anaemia?

Answer

Peak systolic velocity above 1.5 multiples of the median.

#390 · Obstetrics and Gynaecology

How are oligohydramnios and polyhydramnios defined by deepest vertical pocket?

Answer

Oligohydramnios is under 2 cm and polyhydramnios is over 8 cm.

#391 · Obstetrics and Gynaecology

How is fetal ventriculomegaly graded by atrial width?

Answer

Mild 10 to 12 mm, moderate 13 to 15 mm, severe above 15 mm.

#392 · Obstetrics and Gynaecology

What are the lemon and banana signs?

Answer

Frontal bone scalloping and a cerebellum wrapped around the brainstem with an effaced cisterna magna, both indicating open spina bifida.

#393 · Obstetrics and Gynaecology

What distinguishes gastroschisis from omphalocele?

Answer

Gastroschisis is a right paraumbilical defect with free floating bowel and no membrane. Omphalocele is midline with a covering membrane and more often has chromosomal associations.

#394 · Obstetrics and Gynaecology

What is the typical ultrasound picture of left congenital diaphragmatic hernia?

Answer

Stomach bubble in the chest, heart displaced to the right and an absent normal stomach in the abdomen.

#395 · Obstetrics and Gynaecology

Why can a normal four chamber view miss serious cardiac defects?

Answer

Conotruncal defects such as tetralogy of Fallot and transposition are seen on outflow tract views, not the four chamber view.

#396 · Obstetrics and Gynaecology

What defines placenta praevia and a low lying placenta?

Answer

Praevia covers the internal os. Low lying has its edge within 20 mm of the os without covering it.

#397 · Obstetrics and Gynaecology

Which ultrasound signs suggest placenta accreta spectrum?

Answer

Loss of the retroplacental clear zone, irregular intraplacental lacunae, myometrium thinned below 1 mm and bridging vessels to the bladder.

#398 · Obstetrics and Gynaecology

What does the lambda sign indicate?

Answer

A dichorionic twin pregnancy: chorionic tissue extends as a triangle into the base of the intertwin membrane.

#399 · Obstetrics and Gynaecology

What does the T sign indicate?

Answer

A monochorionic diamniotic twin pregnancy, where a thin membrane meets the placenta at a right angle.

#400 · Obstetrics and Gynaecology

What is the main risk of monochorionic monoamniotic twins?

Answer

Cord entanglement, because no dividing membrane is present.

#401 · Obstetrics and Gynaecology

What are Quintero stage I and stage II twin to twin transfusion?

Answer

Stage I is oligohydramnios in the donor (pocket under 2 cm) with polyhydramnios in the recipient (over 8 cm). Stage II adds a donor bladder that is not visible.

#402 · Obstetrics and Gynaecology

Which Quintero stages involve abnormal Doppler, hydrops and demise?

Answer

Stage III abnormal Doppler, stage IV hydrops, stage V demise of one or both twins.

#403 · Obstetrics and Gynaecology

How is a septate uterus distinguished from a bicornuate uterus?

Answer

A septate uterus has a convex or flat external fundal contour. A bicornuate uterus has a fundal cleft of more than 1 cm.

#404 · Obstetrics and Gynaecology

What is the MRI signal of a fibrous uterine septum?

Answer

Low signal on T2 weighted images, whereas a muscular septum is intermediate like myometrium.

#405 · Obstetrics and Gynaecology

What association should be checked when a Mullerian anomaly is found?

Answer

Renal tract anomalies, including renal agenesis, as in OHVIRA (Herlyn Werner Wunderlich) syndrome.

#406 · Obstetrics and Gynaecology

What are the MRI features of red degeneration of a fibroid?

Answer

A high T1 signal rim with absent enhancement, typically in pregnancy.

#407 · Obstetrics and Gynaecology

Which features raise concern for leiomyosarcoma over a fibroid?

Answer

Rapid growth, irregular margin, heterogeneous T2 signal with necrosis, restricted diffusion and T1 bright haemorrhage.

#408 · Obstetrics and Gynaecology

What junctional zone thickness on MRI suggests adenomyosis?

Answer

More than 12 mm, often with T2 bright foci.

#409 · Obstetrics and Gynaecology

How is an endometrioma characterised on MRI and how is it separated from a dermoid?

Answer

T1 bright, stays bright on fat suppression, with T2 shading. A dermoid is also T1 bright but loses signal on fat suppression.

#410 · Obstetrics and Gynaecology

What are the five O-RADS US risk categories?

Answer

1 normal premenopausal, 2 almost certainly benign (under 1 percent), 3 low (1 to under 10), 4 intermediate (10 to under 50), 5 high (50 percent or more).

#411 · Obstetrics and Gynaecology

What is the key Doppler caveat in ovarian torsion?

Answer

Normal arterial and venous flow does not exclude torsion because of the dual ovarian and uterine blood supply. The whirlpool sign is highly specific.

#412 · Obstetrics and Gynaecology

What does an intact low signal cervical stromal ring on T2 MRI imply?

Answer

It excludes parametrial invasion with a very high negative predictive value.

#413 · Obstetrics and Gynaecology

How is FIGO 2018 stage IB subdivided by tumour size?

Answer

IB1 under 2 cm, IB2 2 to under 4 cm, IB3 4 cm or more.

#414 · Obstetrics and Gynaecology

How is FIGO 2009 endometrial cancer stage I divided?

Answer

IA is confined to the endometrium or invades under half the myometrium. IB invades half or more.

#415 · Obstetrics and Gynaecology

How do complete and partial hydatidiform moles differ on ultrasound?

Answer

Complete mole: snowstorm mass, no fetus, theca lutein cysts, very high hCG. Partial mole: triploid, thickened cystic placenta with an abnormal fetus or embryo.

#416 · Paediatrics

What is the diagnostic sign of necrotising enterocolitis on an abdominal radiograph?

Answer

Pneumatosis intestinalis, bubbly or linear lucency in the bowel wall. Portal venous gas indicates more severe disease and pneumoperitoneum indicates perforation.

#417 · Paediatrics

How does transient tachypnoea of the newborn differ from respiratory distress syndrome on a chest film?

Answer

Transient tachypnoea shows normal or hyperinflated lungs with perihilar streaking and fissural fluid, and clears in 48 to 72 hours. Respiratory distress syndrome shows low volumes and fine granular opacity with air bronchograms.

#418 · Paediatrics

Which lung pattern and complication characterise meconium aspiration?

Answer

Hyperinflation with patchy coarse opacities in a term or post term infant. Pneumothorax is a frequent complication.

#419 · Paediatrics

What are the typical findings of a left congenital diaphragmatic hernia on a neonatal chest film?

Answer

Bowel loops in the left hemithorax, rightward mediastinal shift and a paucity of abdominal gas. Pulmonary hypoplasia determines outcome.

#420 · Paediatrics

Where should the tip of an umbilical venous catheter lie?

Answer

At the junction of the inferior vena cava and right atrium. An umbilical arterial catheter lies high at T6 to T10 or low at L3 to L4.

#421 · Paediatrics

What does a double bubble sign indicate and what is it associated with?

Answer

Duodenal atresia, with gas in the stomach and duodenal cap and no distal gas. It is associated with trisomy 21 and annular pancreas.

#422 · Paediatrics

What distinguishes meconium ileus on imaging and what is the underlying disease?

Answer

Dilated loops with a bubbly ground glass appearance in the right lower quadrant and a microcolon on contrast enema. The underlying disease is cystic fibrosis.

#423 · Paediatrics

Which view best shows free gas in a sick neonate with suspected perforation?

Answer

A cross table lateral or left lateral decubitus film, where free gas sits over the liver edge.

#424 · Paediatrics

What are the ultrasound criteria for hypertrophic pyloric stenosis?

Answer

Pyloric muscle thickness of 3 mm or more and canal length of about 15 to 17 mm or more, with a target sign on transverse view.

#425 · Paediatrics

What biochemical disturbance follows prolonged vomiting from pyloric stenosis?

Answer

Hypochloraemic, hypokalaemic metabolic alkalosis.

#426 · Paediatrics

In which age group and sex is pyloric stenosis most typical?

Answer

Infants aged 2 to 8 weeks, more often firstborn boys, with projectile non bilious vomiting.

#427 · Paediatrics

A baby has bilious vomiting. Which test is required even if the plain film is normal?

Answer

An upper gastrointestinal contrast study to look for malrotation and midgut volvulus, because a normal radiograph does not exclude them.

#428 · Paediatrics

Where does the normal duodenojejunal flexure lie on a frontal contrast study?

Answer

To the left of the left pedicle of the vertebral body, at the level of the duodenal bulb. In malrotation it lies low or to the right.

#429 · Paediatrics

What is the whirlpool sign on ultrasound?

Answer

The superior mesenteric vein winding around the superior mesenteric artery, indicating midgut volvulus. A normal scan does not exclude malrotation.

#430 · Paediatrics

What are the ultrasound signs of intussusception?

Answer

The target or doughnut sign on transverse view and the pseudokidney sign on longitudinal view.

#431 · Paediatrics

When should a lead point be suspected in intussusception?

Answer

In children older than 3 years and in small bowel intussusception. Examples are Meckel diverticulum, polyp, lymphoma and Henoch Schonlein purpura haematoma.

#432 · Paediatrics

What are the contraindications to enema reduction of intussusception?

Answer

Perforation and peritonitis.

#433 · Paediatrics

What is the pathological basis of Hirschsprung disease?

Answer

Absent ganglion cells in the distal bowel, producing a narrow aganglionic segment with a transition zone and proximal dilatation.

#434 · Paediatrics

What does the rectosigmoid index show in Hirschsprung disease?

Answer

The ratio of rectal to sigmoid diameter, normally above 1, is reversed. A delayed film at 24 hours shows retained contrast.

#435 · Paediatrics

What is the thymic sail sign and why does it matter?

Answer

The right thymic lobe elevated by the horizontal fissure in an infant. It is normal, does not displace the airway and should not be mistaken for a mediastinal mass.

#436 · Paediatrics

What chest radiograph finding suggests an inhaled foreign body in a toddler?

Answer

Unilateral air trapping: one hyperinflated lung that does not deflate on expiration or decubitus views, with mediastinal shift away on expiration.

#437 · Paediatrics

What are the football sign and Rigler sign?

Answer

Signs of large volume pneumoperitoneum: an oval lucency over the abdomen (football) and both sides of the bowel wall outlined by gas (Rigler).

#438 · Paediatrics

Which barium findings characterise hypertrophic pyloric stenosis?

Answer

The string sign, double track sign and shoulder sign. Ultrasound is the test of choice and shows the target and nipple signs.

#439 · Paediatrics

What is the most common cause of significant neonatal hydronephrosis?

Answer

Pelviureteric junction obstruction, with dilated pelvis and calyces and a normal ureter.

#440 · Paediatrics

Which ultrasound and contrast findings define posterior urethral valves?

Answer

A thick walled bladder with bilateral hydroureteronephrosis and a dilated posterior urethra. The micturating cystourethrogram shows an elongated posterior urethra.

#441 · Paediatrics

How is grade V vesicoureteric reflux described?

Answer

Gross dilatation with tortuous ureters, loss of papillary impressions and intrarenal reflux on the micturating cystourethrogram.

#442 · Paediatrics

Which nuclear medicine test detects renal cortical scarring in children?

Answer

Tc99m DMSA, performed some months after infection to avoid acute changes.

#443 · Paediatrics

How can Wilms tumour be distinguished from neuroblastoma on cross sectional imaging?

Answer

Wilms tumour arises from the kidney with a claw sign and displaces vessels. Neuroblastoma is often calcified, encases vessels, may cross the midline and takes up MIBG.

#444 · Paediatrics

What must a Wilms tumour report comment on beyond the mass itself?

Answer

Renal vein and inferior vena cava extension, contralateral disease and lung metastases.

#445 · Paediatrics

What is the Graf alpha angle threshold for a mature infant hip?

Answer

60 degrees or more. Angles below 43 degrees indicate a decentred hip.

#446 · Paediatrics

How does slipped capital femoral epiphysis appear on imaging?

Answer

Widened physis, posterior and medial slip of the epiphysis, and Klein line failing to intersect the epiphysis. The frog leg lateral view shows it best.

#447 · Paediatrics

What is the crescent sign in Perthes disease?

Answer

A subchondral fracture line in the femoral epiphysis, marking the stage before fragmentation.

#448 · Paediatrics

What radiographic features indicate rickets?

Answer

Widened physes with cupped, frayed metaphyses, most obvious at the wrist and knee, with poorly mineralised epiphyses.

#449 · Paediatrics

What is a toddler fracture?

Answer

An oblique or spiral tibial shaft fracture in a child learning to walk, which may be seen only on an oblique view.

#450 · Paediatrics

Which fractures have high specificity for non accidental injury?

Answer

Classic metaphyseal lesions, posterior rib fractures, scapular, sternal and spinous process fractures, and multiple fractures of different ages.

#451 · Paediatrics

Why is a skeletal survey repeated about two weeks later in a child under two?

Answer

To detect healing injuries and fractures that were occult initially.

#452 · Paediatrics

How is germinal matrix haemorrhage graded on cranial ultrasound?

Answer

Papile grades I to IV: matrix only, intraventricular without dilatation, with ventricular dilatation, then intraparenchymal haemorrhage.

#453 · Paediatrics

Which childhood posterior fossa tumour is hyperdense on CT, arises from the vermis and spreads via cerebrospinal fluid?

Answer

Medulloblastoma, which shows restricted diffusion and needs whole spine imaging.

#454 · Paediatrics

What is the typical appearance of cerebellar pilocytic astrocytoma?

Answer

A cystic mass with an enhancing mural nodule in a cerebellar hemisphere, without restricted diffusion.

#455 · Paediatrics

What are the key approaches to reducing CT dose in children?

Answer

Justify the test, use size based protocols with lower tube current and voltage, limit scan range, avoid multiphase scans and use iterative reconstruction.

#456 · Interventional and Multisystem

Which agents are used for permanent distal embolisation in fibroids and prostate artery embolisation?

Answer

Polyvinyl alcohol particles and calibrated microspheres reach the arteriolar bed and give a permanent effect.

#457 · Interventional and Multisystem

Which embolic agent is temporary and suits trauma and postpartum haemorrhage?

Answer

Gelatin sponge, which lasts days to weeks.

#458 · Interventional and Multisystem

What is the dangerous complication to exclude before bronchial artery embolisation?

Answer

Spinal cord infarction from a radiculomedullary branch supplying the anterior spinal artery; look for the anterior spinal blush first.

#459 · Interventional and Multisystem

What is the usual source of massive haemoptysis treated by embolisation?

Answer

The bronchial arteries.

#460 · Interventional and Multisystem

What is post-embolisation syndrome?

Answer

Pain, fever and malaise that settle within days. It is expected and does not indicate infection.

#461 · Interventional and Multisystem

Which route does a TIPS usually take?

Answer

From the right hepatic vein to the right portal vein through the liver, using a covered stent.

#462 · Interventional and Multisystem

What are the main indications for TIPS?

Answer

Variceal bleeding refractory to endoscopic and medical therapy, and refractory ascites.

#463 · Interventional and Multisystem

What is the main complication of TIPS and why?

Answer

Hepatic encephalopathy, because portal blood bypasses the liver.

#464 · Interventional and Multisystem

Name contraindications to TIPS.

Answer

Right heart failure, severe pulmonary hypertension, uncontrolled sepsis and unrelieved biliary obstruction; polycystic liver is relative.

#465 · Interventional and Multisystem

What Doppler finding suggests TIPS dysfunction?

Answer

Low or absent stent flow, loss of the high shunt velocity and return of slow or hepatofugal portal flow.

#466 · Interventional and Multisystem

What is the correct first step for an infected obstructed kidney?

Answer

Percutaneous nephrostomy to decompress the system, with stone treatment delayed until sepsis settles.

#467 · Interventional and Multisystem

When is the Seldinger technique preferred over trocar drainage?

Answer

Seldinger places the catheter over a wire and suits deeper or smaller collections; trocar suits large superficial collections.

#468 · Interventional and Multisystem

What bleeding thresholds are used for high-risk percutaneous procedures?

Answer

Roughly INR 1.5 to 1.8 or lower and platelets above about 50 x 10^9 per litre, subject to local protocol.

#469 · Interventional and Multisystem

What is the commonest complication of CT-guided lung biopsy?

Answer

Pneumothorax, more likely with emphysema, a deep lesion and a long needle path.

#470 · Interventional and Multisystem

What is the heat sink effect?

Answer

Flowing blood in an adjacent vessel carries heat away during radiofrequency ablation, leaving a viable rim of tumour.

#471 · Interventional and Multisystem

Which ablation method produces an ice ball visible on CT and MRI?

Answer

Cryoablation.

#472 · Interventional and Multisystem

What size of hepatocellular carcinoma is the best target for curative ablation?

Answer

A solitary lesion up to about 3 cm.

#473 · Interventional and Multisystem

How do you separate benign hyperaemia from residual tumour after ablation?

Answer

Nodular or crescentic enhancement at the margin is residual tumour; a thin uniform smooth rim in the first month is benign.

#474 · Interventional and Multisystem

What is the principal indication for an IVC filter?

Answer

Acute proximal DVT or pulmonary embolism when anticoagulation is contraindicated or has failed.

#475 · Interventional and Multisystem

Where is an IVC filter usually placed, and when higher?

Answer

Infrarenal in most patients; suprarenal if thrombus extends to the renal veins or in pregnancy.

#476 · Interventional and Multisystem

List complications of IVC filters.

Answer

Migration, strut fracture and embolisation, caval perforation, caval thrombosis and tilting that hinders retrieval.

#477 · Interventional and Multisystem

What is a type I endoleak?

Answer

A leak at the proximal or distal graft attachment site; high pressure, rupture risk, needs prompt treatment.

#478 · Interventional and Multisystem

What is a type II endoleak and how is it managed?

Answer

Retrograde flow from a lumbar artery or inferior mesenteric artery. The commonest type; observe unless the sac enlarges, then embolise.

#479 · Interventional and Multisystem

What is a type III endoleak?

Answer

A leak through a graft defect or between overlapping graft components; high pressure and needs prompt treatment.

#480 · Interventional and Multisystem

What is a type IV endoleak?

Answer

Leak through porous graft fabric seen early after placement; self-limiting.

#481 · Interventional and Multisystem

What is endotension (type V)?

Answer

Sac enlargement with no demonstrable leak on imaging.

#482 · Interventional and Multisystem

Which imaging phase best separates type I from type II endoleak?

Answer

Type I appears in the arterial phase near the graft ends; type II appears on the delayed phase from a branch vessel.

#483 · Interventional and Multisystem

What ankle brachial index indicates peripheral arterial disease, and when is it unreliable?

Answer

Below 0.9 indicates disease; above 1.3 means calcified incompressible vessels, so use toe pressure or waveforms.

#484 · Interventional and Multisystem

What peak systolic velocity ratio suggests a 50 percent or greater stenosis on duplex?

Answer

A ratio above 2; above 4 suggests 75 percent or more.

#485 · Interventional and Multisystem

What is May-Thurner syndrome?

Answer

Compression of the left common iliac vein by the right common iliac artery, causing left-sided iliofemoral DVT.

#486 · Interventional and Multisystem

What is the first-line treatment of anaphylaxis to contrast?

Answer

Intramuscular adrenaline, 0.5 mg of 1 in 1000 in an adult, with airway support, oxygen and fluids.

#487 · Interventional and Multisystem

Who carries the highest risk of contrast-associated kidney injury?

Answer

Patients with eGFR below about 30, diabetes, dehydration or intra-arterial administration.

#488 · Interventional and Multisystem

Which gadolinium agents are preferred when eGFR is below 30?

Answer

Group II agents, mostly macrocyclic, which carry the lowest risk of nephrogenic systemic fibrosis.

#489 · Interventional and Multisystem

Where does gadolinium deposit in the brain?

Answer

The dentate nucleus and globus pallidus, mostly after repeated linear agent exposure, with no established harm.

#490 · Interventional and Multisystem

What is the occupational whole-body dose limit?

Answer

20 mSv per year averaged over five years, with no single year above 50 mSv.

#491 · Interventional and Multisystem

Name the three principles of radiation protection.

Answer

Justification, optimisation (as low as reasonably achievable) and dose limitation.

#492 · Interventional and Multisystem

At what dose does transient skin erythema occur?

Answer

About 2 Gy, with epilation at about 3 Gy; these are deterministic effects.

#493 · Interventional and Multisystem

Which optic and brain tumours characterise NF1 and NF2?

Answer

NF1: optic pathway glioma and neurofibromas.
NF2: bilateral vestibular schwannomas, meningiomas and ependymomas.

#494 · Interventional and Multisystem

Which tuberous sclerosis findings occur in the kidney, heart and brain?

Answer

Renal angiomyolipomas, infantile cardiac rhabdomyomas, and cortical tubers with subependymal nodules or giant cell astrocytoma.

#495 · Interventional and Multisystem

What are the features of Von Hippel-Lindau disease?

Answer

Cerebellar and retinal haemangioblastomas, clear cell renal carcinoma, phaeochromocytoma and pancreatic cysts or neuroendocrine tumours.

#496 · Interventional and Multisystem

What is the sarcoidosis 1-2-3 sign?

Answer

Bilateral hilar plus right paratracheal lymphadenopathy, symmetrical in distribution.

#497 · Interventional and Multisystem

What are the Deauville scores 1 to 5?

Answer

1 no uptake, 2 at or below mediastinum, 3 above mediastinum up to liver, 4 moderately above liver, 5 markedly above liver or new lesions.

#498 · Interventional and Multisystem

What imaging appearance suggests autoimmune pancreatitis rather than pancreatic cancer?

Answer

A diffusely enlarged sausage-shaped gland with a delayed-enhancing capsule-like rim that responds to steroids, with raised serum IgG4.

#499 · Interventional and Multisystem

Which vasculitis causes coronary aneurysms in children and which causes renal microaneurysms?

Answer

Kawasaki disease causes coronary aneurysms; polyarteritis nodosa causes renal and mesenteric microaneurysms.

#500 · Interventional and Multisystem

Name common causes of false-positive and false-negative FDG PET-CT.

Answer

False positives: infection, sarcoidosis, recent treatment. False negatives: mucinous or low-grade tumours, lesions under about 8 mm, hyperglycaemia.